AIMILIOS ROMAIOS

Mechanical Engineer

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Professional Summary

Mechanical Engineer with a strong technical background in numerical simulation, CFD, fluid mechanics and CAD. Experienced in developing 3D models and applying simulation-driven methods to analyse complex physical behaviour, with hands-on experience in SolidWorks Flow Simulation and Fusion 360. Strong interest in expanding into structural, thermal and multiphysics modelling, combining analytical thinking, numerical methods and engineering design to solve challenging engineering problems.

Education

Master's Degree in Mechanical Engineering – Grade: 8.22
University of Peloponnese, Patra, Greece – 09/2019 – 01/2025

- Computational Fluid Dynamics (CFD) and fluid mechanics
- CAD design and mechanical systems
- Engineering simulations and numerical analysis
- Automotive engineering and advanced manufacturing concepts
- Design, optimization, and mechanical performance analysis

Master's Thesis / Engineering Project:
Aerodynamic Analysis of F1 Race Car with Computational Fluid Dynamics (CFD)

- Studied fundamental principles of vehicle aerodynamics and their impact on race car performance
- Analyzed equations related to fluid flow and aerodynamic behavior
- Designed a 3D Formula 1 Race Car model using Fusion 360
- Performed CFD simulations using SolidWorks Flow Simulation to evaluate aerodynamic characteristics and optimize performance
- Investigated aerodynamic parameters including airflow behavior, pressure distribution, and performance efficiency

Work Experience

Assistant Inspector (Internship)
KTEO Ioniki S.A., Pyrgos, - 09/2024 - 11/2024

- Performed vehicle identification and registration verification procedures.
- Conducted exhaust emission testing for petrol and diesel vehicles.
- Operated inspection equipment including brake tester, suspension tester, and camber measurement tools.

Motorsport Columnist
f1fan.gr | Remote - 01/2026 – Present

- Write technical motorsport articles and analysis for an online audience
- Develop knowledge of modern motorsport technologies and vehicle development

Stock Manager
AFOI TERZI S.A. – Electrical Materials
Pyrgos, Greece - 05/2026 – 09/2026

- Managing inventory operations, including receiving, organizing, and tracking electrical materials and equipment
- Maintain accurate stock records and monitor product availability to support daily operations
- Coordinate material handling, storage organization, and order preparation

Engineering Projects

Hyperion Drone Team
University of Peloponnese, Patra, Greece
Team Member | 11/2024 – 11/2025

- Designed and optimized drone configurations with a focus on aerodynamic efficiency and performance
- Created prototypes using 3D printing technologies and supported the manufacturing process
- Applied CAD modeling and engineering design principles to develop and improve components
- Collaborated with a multidisciplinary team to test, evaluate, and refine design solutions

Certifications

- NOCN Level 3 Certificate in ESOL International (C2 Proficiency)
- Certificate of Participation – Mastering Simulation and Data Science with Altair
    - Introduction to simulation-driven engineering and data science applications using Altair technologies
- Basic Office Skills Certificate
- MATLAB Onramp Course Completion Certificate

Skills and Qualities

Technical Skills
CAD & Engineering Design

- Advanced CAD modeling and mechanical design using Fusion 360, SolidWorks, and AutoCAD.
- 3D modeling, design optimization, and preparation of components for prototyping.

CFD & Simulation

- CFD simulation and aerodynamic analysis using ANSYS Fluent, SolidWorks Flow Simulation, SimScale, and OpenFOAM
- Aerodynamic performance evaluation, including airflow analysis, drag reduction, and downforce optimization

CAE & Prototyping

- FEA fundamentals for structural analysis
- 3D printing, rapid prototyping, and physical model validation

Additional Skills

- Technical drawing and automotive sketching with strong attention to detail
- Graphic design skills using Adobe Photoshop for visual content creation and technical presentations
- Strong teamwork and collaboration skills developed through engineering projects and multidisciplinary teams
- Analytical thinking and creative problem-solving approach for engineering challenges
- High adaptability and ability to learn new tools, technologies, and workflows quickly
- Excellent organization skills with accuracy and focus on quality results
- Ability to work effectively under pressure and meet demanding deadlines

Military Service

Andravida, Ilia – 02/2025 - 02/2026
Specialty: Driver (Follow Me)